The Signaling Gap

State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine

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The Puzzle

When Formal Channels Become Liabilities

States that depend on a powerful patron for their security **do not always have the luxury of saying what they think.**

- Formal channels (UN votes, official statements) risk retaliation from the patron
- But **silence risks being seen as endorsement** when the patron draws international condemnation
- General problem: any hierarchical relationship where a subordinate depends on a patron but must also balance other audiences

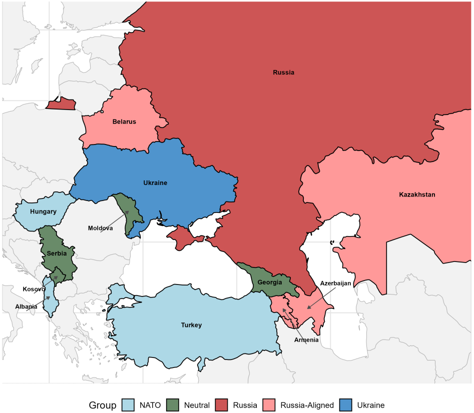
Question: What channels do constrained states use to signal genuine preferences, and when are they credible?

February 24, 2022

The Russian invasion of Ukraine forced Russia's security partners to navigate this bind.

- CSTO members had genuine reasons to oppose the invasion
- Violated sovereignty norms that protect small states
- Threatened substantial Western trade ties
- But formal condemnation of a patron willing to use force → not viable

Study Region: Russia's Near Abroad



A Gap Between Scholars and Practitioners

IR Scholarship

State media studied as a **domestic** tool:

- Agenda setting (McCombs 2002)
- Spin dictators (Guriev & Treisman 2022)
- Autocratic media management (Egorov et al. 2009)

Orientation: how governments shape what *their own publics* think

Intelligence Practice

State media monitored as **international signal**:

- BBC Monitoring (est. 1939)
- FBIS / Kremlinology (Cold War)
- OSINT formalized (Manor 2019; USIC 2024)

Orientation: reading state media as a window into *regime preferences*

Same outlet does both. Tools exist — but haven't been assembled.

This Paper

Argument: State-owned media provides an informal signaling channel through which constrained states communicate genuine preferences to external audiences.

Preview of findings:

1. The invasion produced a significant negative shift in media sentiment toward Russia across the region
2. State-owned media shifted *less* negatively than independent media → strategic management
3. State media in Russia-*aligned* states shifted *more* negatively than neutral states — **opposite of domestic suppression prediction**

These states refused to formally condemn the invasion while allowing their state media to turn sharply critical.

Theory

Signaling Under Constraint

Standard signaling theory: credible signals require costs (Fearon 1997)

- Audience costs: leaders stake political capital on public positions
- Sunk costs: troop deployments, sanctions, arms transfers
- Common feature: **visible, attributable, formal** channels

But alliances constrain:

- Alliance membership trades autonomy for security (Morrow 1991)
- Public dissent risks abandonment (Snyder 1997)
- Formal channels lock precisely when dissent is most needed (Weitsman 2004)

$\rightarrow$ **What do constrained states do when formal channels are unavailable?**

The Signaling Mechanism

State media signals work through two properties:

1. **Readable**
   - Common knowledge of state control $\rightarrow$ deviation from baseline is informative (Gehlbach et al. 2014)
   - Foreign analysts read shifts as regime choice, not journalistic noise
2. **Costly**
   - Allowing negative coverage of a security patron risks provoking retaliation from Moscow
   - A state with no genuine objection would not accept this risk $\rightarrow$ bluffing is irrational

But the Cost is Bounded

Key insight: audience-specific visibility

Channel	Western Intel	Moscow	Russian Public
UN vote against Russia	✓	✓	✓
Official diplomatic statement	✓	✓	Likely
State media shift	✓	**If looking**	×
Private diplomacy	Unverifiable	Possible	×

- Authoritarian survival → patrons most responsive to signals visible to their *winning coalition* (Bueno de Mesquita et al. 2003; Guriev & Treisman 2022)
- Kazakh state media reaches Western analysts but is invisible to the Russian public
- State media operates **below the patron's domestic visibility threshold**

Weak Deniability as a Secondary Off-Ramp

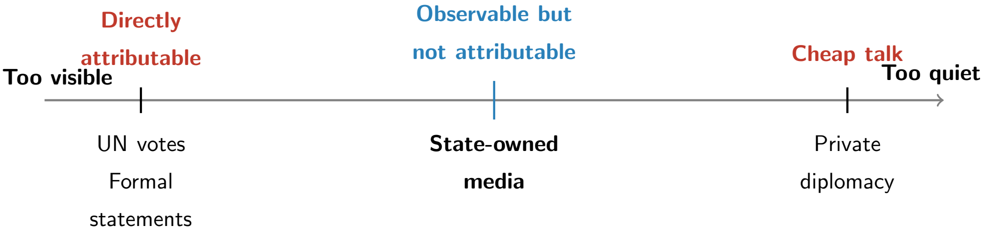
If confronted: institutional separation between government and state media provides a fallback

- Coverage can be disavowed as editorial judgment rather than state policy (Gehlbach et al. 2014, Entman 2004)
- This preserves the patron relationship even as the signal reaches its intended audience

Note: This is not the core mechanism — it is a secondary feature that makes the channel *sustainable*.

Core logic:
Signal is **costly** (risk of retaliation) $\rightarrow$ but cost is **bounded** (audience-specific visibility) $\rightarrow$ with **weak deniability** as a fallback

Why State Media Rather Than Alternatives?



- Public enough to be observed by foreign governments that systematically monitor it
- Quiet enough to operate below the threshold of provoking a response from Moscow
- Credible because external observers know state media reflects regime preferences

Hypotheses

H1: The invasion produced a significant decline in media sentiment toward Russia across both state-owned and independent outlets.

H2: State-owned media will report negative sentiment towards Russia after the invasion to a *lesser* extent than independent media.

H3: State-owned media in states with formal security agreements with Russia will respond *more negatively* to the invasion than state-owned media in neutral states.

Competing Predictions for H3

Domestic Suppression

- Russia-aligned states **suppress** negative coverage to protect alliance
- State media *less* negative than neutral states
- Alternative: Higher ethnic Russian pop. → **more** negative

Signaling

- Russia-aligned states use state media to **signal dissent** because formal channels are foreclosed
- State media *more* negative than neutral states
- Ethnic Russian pop. → **no effect** (logic is security alignment, not demographics)

Opposite predictions provide leverage for distinguishing between mechanisms.

Data

Sample Overview

Coverage: Aug 2021 – Aug 2022
Articles: ~174,000 (excl. Ukraine)
Observation: Publication Date-Time
Sources: 62 outlets, 11 countries
Languages: Translated via HuggingFace

Russia-Aligned	Armenia, Azerbaijan, Belarus, Kazakhstan
Neutral	Georgia, Moldova, Kosovo, Serbia
NATO	Albania, Hungary, Turkey

Ethnic Russian Pop. × Security Alignment

	Neutral	**CSTO/Aligned**
Low	Kosovo, Serbia	Azerbaijan, Armenia
High	Moldova, Georgia	Belarus, Kazakhstan

Orthogonal variation enables separate tests of signaling vs. domestic audience mechanisms.

Measurement: GPT-4o Classification

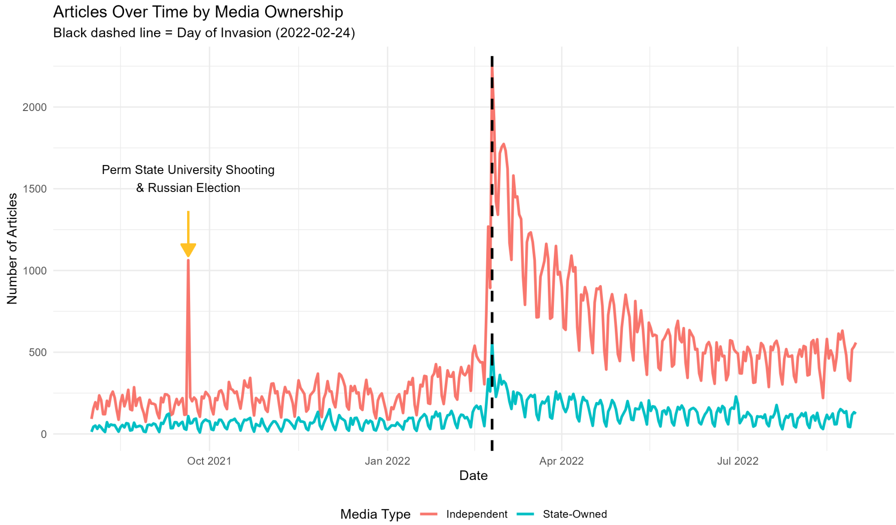
Three dimensions coded at article level (title + first two sentences):

1. **Topic:** Political, Economic, Military, Cultural, Other
2. **Russia involvement:** Binary (main actor or not)
3. **Sentiment toward Russia:** Ternary (-1, 0, $+1$)

Validation: 100 adversarially double-coded articles (coauthors debated disagreements)

Dimension	Initial Human Agreement	GPT F1 Score
Sentiment	79%	0.92
Russia involvement	91%	0.83
Topic	93%	0.80

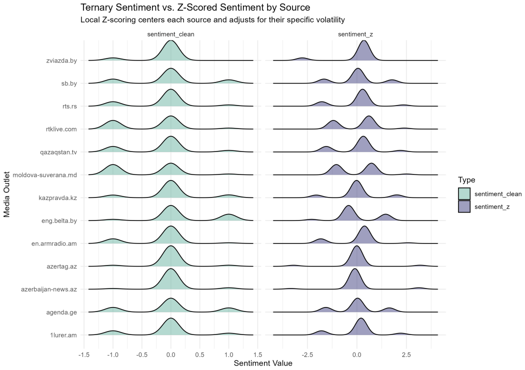
Article Volume Over Time



Source-Level Z-Score Normalization

Problem: Outlets have different baseline sentiment levels and volatility.

Solution: Z-score normalizes each source to its own pre-invasion distribution.



Empirical Strategy

Identification Strategy

Core approach: Regression Difference-in-Discontinuity in Time

- February 24 as a sharp break in the information environment
- Troop buildup was gradual; full-scale invasion was discrete and unanticipated in scale
- Running variable: "Probabilistic" Publication Time (measured at date-time)

Within-group RDDs are causal; cross-group comparisons are correlational but subject to extensive robustness checks. Where missing, publication time is replaced by a source or country level KDE. Findings robust to alternative time specifications

Three Levels of Analysis

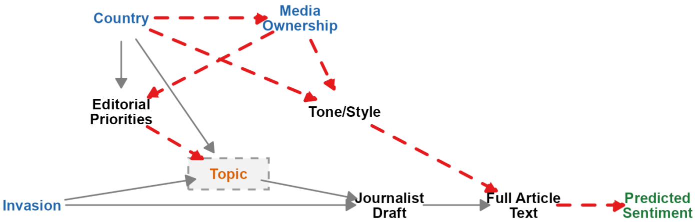
1. **H1 — RDD:** Effect of invasion on sentiment (all media)
2. **H2 — Diff-in-Disc:** State-owned $\times$ Treatment interaction
3. **H3 — Diff-in-Disc:** CSTO $\times$ Treatment interaction (state media only)

Key design choices:

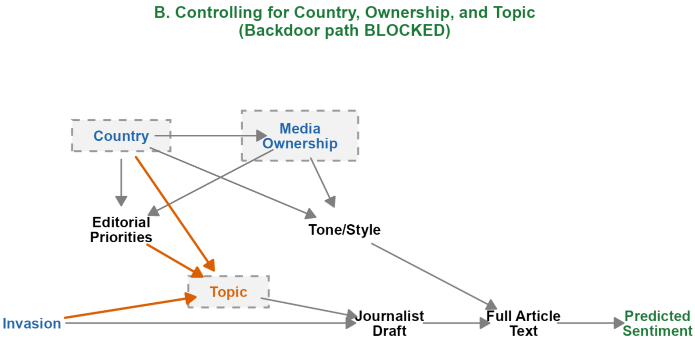
- Bandwidth selection via `rdrobust` (Calonico, Cattaneo & Titiunik)
- Source-level Z-score normalization or source/state/topic fixed effects
- Clustering at the source level; wild cluster bootstrap for few-cluster settings

The Identification Problem

A. Controlling for Topic Only
(Backdoor path OPEN)



The Solution: Conditioning on Country, Ownership, Topic



Specification: H1 (RDD)

$$Y_i = \alpha + \boldsymbol{\tau} D_i + \beta_1 X_i + \beta_2 (D_i \times X_i) + Z_i \gamma + \varepsilon_i$$

Y_i	Ternary or Z-Score Normalized (at source level) Sentiment
D_i	Post-invasion indicator
X_i	Running variable (probabilistic date-time)
Z_i	Topic FEs
$\boldsymbol{\tau}$	**Coefficient of interest:** causal effect of invasion

Bandwidth selected via `rdrobust` (Calonico, Cattaneo & Titiunik). SEs clustered at source level.

Specification: H2 (Diff-in-Disc)

$$
\begin{aligned}
Y_i = & \alpha + \tau D_i + \beta_1 X_i + \beta_2 (D_i \times X_i) + \beta_3 S_i + \boldsymbol{\delta(S_i \times D_i)} + \\
& \beta_5 (S_i \times X_i) + \beta_6 (S_i \times D_i \times X_i) + Z_i \gamma + \varepsilon_i
\end{aligned}
$$

S_i	State-owned indicator
$\boldsymbol{\delta}$	**Coefficient of interest:** differential treatment effect for state vs. independent media

All models include topic FEs and source-level Z-score normalization. Bandwidth via `rdrobust`. WCR bootstrap for inference ($B = 4{,}999$, Webb weights).

Specification: H3 (Diff-in-Disc, State Media Only)

$$
\begin{aligned}
Y_i = & \alpha + \tau D_i + \beta_1 X_i + \beta_2 (D_i \times X_i) + \beta_3 C_i + \boldsymbol{\delta (C_i \times D_i)} + \\
& \beta_5 (C_i \times X_i) + \beta_6 (C_i \times D_i \times X_i) + Z_i \gamma + \varepsilon_i
\end{aligned}
$$

C_i	CSTO/Russia-aligned indicator
$\boldsymbol{\delta}$	**Coefficient of interest:** differential treatment effect for CSTO vs. neutral state media

Sample restricted to state-owned media in non-NATO countries. Topic FEs + Z-score normalization. Bandwidth via `rdrobust`. WCR bootstrap for inference ($B = 4{,}999$, Webb weights).

Inference with Few Clusters

Challenge: Source-level clustering yields ∼13–60 clusters.

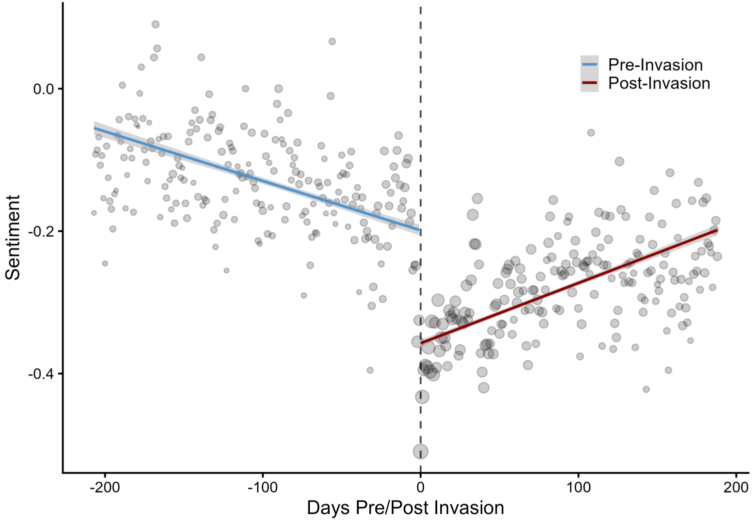
Approach:

- H1: Cluster-robust SEs from `rdrobust` (60 clusters)
- H2/H3 interactions: Wild cluster restricted (WCR) bootstrap — Cameron, Gelbach & Miller (2008), Webb six-point weights, $B = 4{,}999$, null imposed
- Additional: exact matching on topic + publication date

Robustness toolkit: placebo tests, leave-one-out, OVB sensitivity ▸ Details

Results

H1: The Invasion Produced a Negative Sentiment Shock

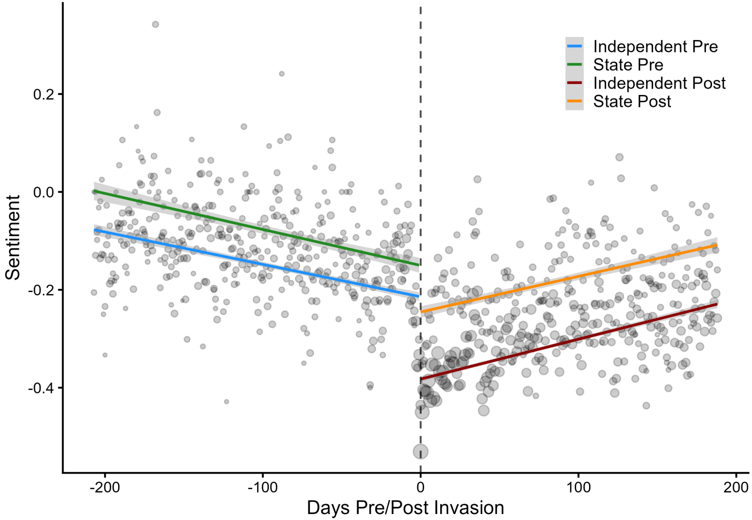


H1: Robust Across Specifications

	(1) Baseline	(2) State + Source FEs	(3) State + Source + Topic FEs	(4) Z-Score Non-Mil.	(5) No NATO
Invasion	−0.130***	−0.119***	−0.106***	−0.184***	−0.284***
	(0.022)	(0.015)	(0.015)	(0.033)	(0.038)

All models use the optimal bandwidth. SEs clustered at source level. Model 4 uses source-level Z-score normalization and omits military topics. Model 5 additionally excludes NATO allies.

H2: State Media Managed Coverage More Carefully



H2: State Ownership Interaction

	(1)‡ State RDD	(2)‡ Indep. RDD	(3)† Non-Mil.	(4)† Non-NATO	(5)† Matched
Invasion	−0.109+	−0.214***	−0.240***	−0.347***	−0.348***
	(0.058)	(0.034)	(0.031)	(0.027)	(0.026)
State × Treat			0.14+	0.222*	0.222*
			(0.06)	(0.069)	(0.069)

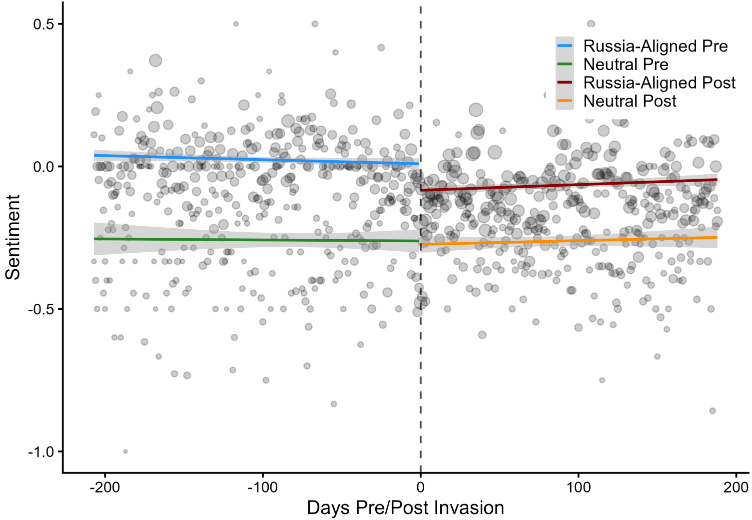
‡Separate RDDs with cluster-robust SEs. †WCR bootstrap ($B = 4{,}999$). (3) omits military topics. (4) omits NATO allies. (5) exact matching on topic + date. All use Z-score normalization and optimal bandwidth.

H2: State Management Varies by Topic

	Political	Military	Economic	Cultural	Other
Invasion	-0.213***	-0.422***	-0.423***	-0.547***	-0.269***
	(0.027)	(0.068)	(0.066)	(0.071)	(0.039)
State $\times$ Treat	0.155	0.012	0.290*	0.100	0.270**
	(0.079)	(0.102)	(0.130)	(0.190)	(0.100)

State management strongest in Economic and Other coverage; not significant for Military.
Consistent with strategic topic selection — military topics are harder to spin.

H3: Russia-Aligned State Media Shifted More Negatively



H3: CSTO Interaction — The Critical Test

	(1)‡ CSTO RDD	(2)‡ Neutral RDD	(3)† Non-Mil.	(4)† Matched
Invasion	−0.220*	0.055	0.098	0.098
	(0.099)	(0.101)	(0.070)	(0.069)
CSTO $\times$ Treat			−0.330*	−0.380*
			(0.110)	(0.110)

‡Separate RDDs with cluster-robust SEs. †WCR bootstrap ($B = 4{,}999$). Sample restricted to state-owned media in non-NATO countries. (3) omits military topics. (4) exact matching on topic + date. All use Z-score normalization and optimal bandwidth.

Ethnic Russian Population: The Null That Matters

	(1)‡ High RDD	(2)‡ Low RDD	(3)† Optimal BW	(4)† Matched
Invasion	−0.079	−0.108	−0.170	−0.143
	(0.066)	(0.102)	(0.100)	(0.099)
High Russian × Treat			−0.055	0.015
			(0.133)	(0.145)

- If domestic audience management were driving the pattern, ethnic Russian population should matter
- **It doesn't** — interaction is small, unstable in sign, never significant
- Rules out the demographic mechanism; supports the signaling interpretation

Putting It Together

Hypothesis	Prediction	Result
H1: Baseline shock	Negative shift	**Confirmed** (-0.13 to -0.28)
H2: State vs. Independent	State less negative	**Confirmed** ($+0.22$ SD)
H3: CSTO vs. Neutral	CSTO *more* negative	**Confirmed** (-0.33 to -0.38 SD)
H3b: Ethnic Russian pop.	No effect	**Confirmed** (null)

- H3 and H3b jointly distinguish signaling from domestic suppression
- Pattern is *opposite* of what audience management predicts
- Robust to bandwidth, matching, topic restrictions, leave-one-out ▸ Details

Discussion

What the Findings Mean

States whose formal channels are constrained by dependence on a powerful patron produce systematically different media signals than states whose channels remain open.

- State media sentiment is not noise — it is a legible instrument of international communication
- The channel works because cost is *bounded* by audience-specific visibility
- The same outlet simultaneously manages domestic audiences (H2) and signals to external ones (H3)

Cohen (1987) was right. The channel Soviet-aligned states used during the Cold War remains operative — and is now empirically identifiable.

Contributions

1. **Bridging the theory-practice gap**
    - Signals practitioners monitored for 80+ years are theoretically structured and empirically identifiable
    - Not idiosyncratic — follows from the logic of alliance constraints
2. **Informal signaling under alliance constraints**
    - When formal channels are foreclosed, states turn to informal ones with predictable patterns
    - General framework: any patron-client relationship under constraint
3. **Methodological**

Limitations and Extensions

Limitations:

- Single case — but theory generalizes to any patron-client relationship under constraint
- Cannot directly observe signal *receipt* — we identify the sent signal, not the received one
- Factional incoherence vs. strategic coordination: RDD identifies the effect regardless of internal mechanism

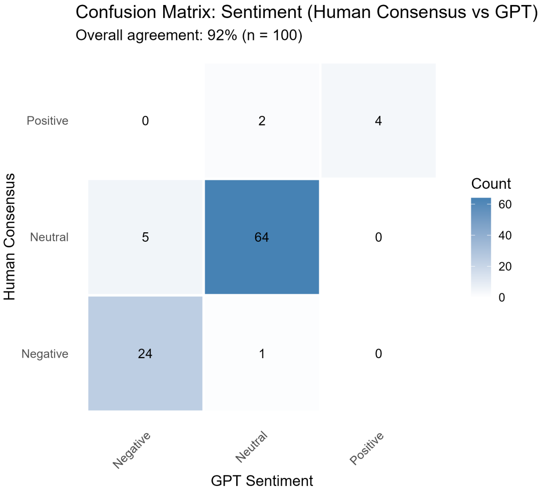
Extensions:

- Other patron-client dyads (U.S.–Central America in the 1980s/90s)
- Temporal dynamics: does signaling evolve as the conflict persists?
- Reception: do policy shifts in Western states correlate with media signals

Thank You

Appendix

GPT Classifier: Sentiment Confusion Matrix



92% overall agreement ($n = 100$). Adversarial double-coding by coauthors with consensus

Robustness Summary

Placebo test: False treatment at Dec 7, 2021 (Putin's announcement) $\rightarrow$ null

Model	Estimate	SE	p
H1: Full RDD	-0.009	(0.031)	0.775
H2: Diff-in-Disc	0.121	(0.056)	0.080
H3: Non-Military	0.705	(0.287)	0.143

Additional checks:

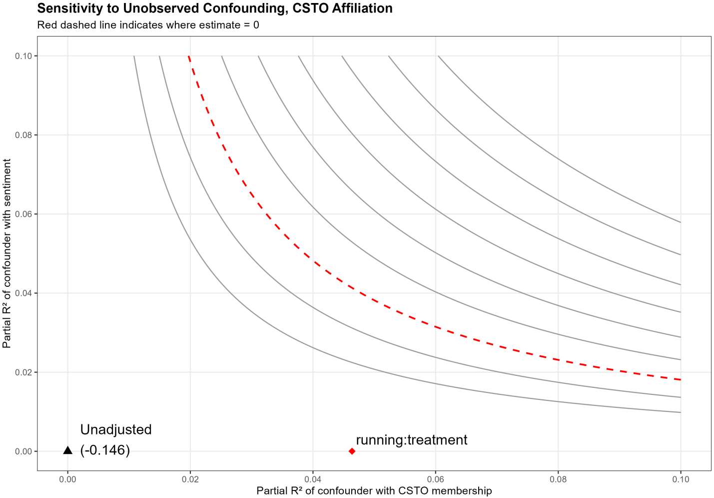
- Leave-one-out by country: survives all exclusions (Belarus attenuates) ▸ LOO
- Exact matching on topic + publication date
- OVB sensitivity analysis ▸ OVB plot
- Multiple bandwidth specifications; drop military topics (H3 strengthens)

H3: Leave-One-Out Country Robustness

	Standard Model				*Matched Model*		
Dropped	**Coeff.**	**SE**	**p**	**Dropped**	**Coeff.**	**SE**	**p**
Belarus	−0.188	(0.136)	0.201	Belarus	−0.203	(0.142)	0.186
Armenia	−0.258*	(0.102)	0.030	Armenia	−0.344*	(0.109)	0.010
Georgia	−0.230*	(0.101)	0.043	Georgia	−0.280*	(0.107)	0.024
Azerbaijan	−0.361*	(0.136)	0.024	Azerbaijan	−0.387*	(0.136)	0.017
Kazakhstan	−0.249*	(0.085)	0.015	Kazakhstan	−0.277**	(0.076)	0.004
Moldova	−0.326*	(0.114)	0.015	Moldova	−0.383**	(0.110)	0.005
Kosovo	−0.298	(0.138)	0.054	Kosovo	−0.348*	(0.130)	0.021
Serbia	−0.387**	(0.112)	0.005	Serbia	−0.407**	(0.112)	0.004

Belarus attenuation expected — as the most constrained state, its removal reduces the signal. Result otherwise stable across all exclusions.

OVB Sensitivity Analysis: CSTO Affiliation



Red dashed line: combinations of confounding that would reduce estimate to zero. Unadjusted

Daily Sentiment Counts by Media Type